ASSIGNMENT 3

Q2 - Employee Bonus Calculator

JavaScript solution

```
// q2.js
"use strict";

const employees = [
  { name: "Amit", salary: "45000", years: "5" },
  { name: "Sara", salary: "38000", years: "2" },
  { name: "Kiran", salary: "52000", years: "7" }
];

for (const employee of employees) {
  try {
    if (!employee || !("name" in employee) ||
        !("salary" in employee) || !("years" in employee)) {
      throw new Error("Missing employee property");
    }

    const salary = Number(employee.salary);
    const years = Number(employee.years);
    if (!Number.isFinite(salary) || !Number.isFinite(years)) {
      throw new Error("Salary or years is not a valid number");
    }

    const bonus = years > 3 ? salary * 0.10 : salary * 0.05;
    console.log(`${employee.name}: Salary = ■${salary.toFixed(2)}, Years = ${years}, Bonus = ■${bonus.toFixed(2)}, Total = ■
  } catch (error) {
    console.log(`Employee record error: ${error.message}`);
  }
}
```

Explanation / Debug notes

The salary and years strings are converted with Number(). Employees with more than 3 years receive 10%; others receive 5%. Strict mode prevents accidental creation of undeclared global variables. try...catch keeps one bad record from stopping the loop.

Run the code in the browser console or Node.js. Save each solution in its own .js file as required.